REPORT

entitled

"CRIME DATA ANALYTICS – CHICAGO CRIME DATASET"

Submitted to the Department of Computer Science and Engineering in partial fulfillment of the requirement for the Degree of

Bachelor of Technology

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1. Problem Description

Crime is not random. Behind every incident report is a pattern, a neighbourhood that sees consistent trouble, a time of night that is statistically more dangerous, a type of offence that clusters around particular locations. The challenge for a large city is turning millions of raw records into something that actually informs decisions.

Chicago maintains one of the richest publicly available crime datasets in the world, over 7.8 million reported incidents spanning more than two decades. This project uses that data to answer four practical questions:

- Which crime types dominate, and how have volumes shifted over time?
- Are there meaningful temporal rhythms: by hour, day, and season, that could shape patrol scheduling?
- Can geographic clustering identify distinct hotspot zones, each with its own crime profile?
- Can a model predict whether a given incident will result in an arrest, and what drives that outcome?

The answers have direct operational value: hotspot maps help allocate limited resources, temporal patterns enable smarter scheduling, and an arrest-prediction model can help investigators prioritise cases.

2. Dataset Description

2.1 Source

The Chicago Crime Dataset is published by the City of Chicago's Open Data Portal. It records every reported crime from 2001 to the present, updated on a near-real-time basis.

URL: https://data.cityofchicago.org/api/views/ijzp-q8t2/rows.csv?accessType=DOWNLOAD

After cleaning, the working dataset contained **7,818,757 records across 19 columns**, spanning 2001 to 2026.

2.2 Key Attributes

Column	Description
ID	Unique identifier for each incident — used for deduplication
Date	Date and time of the incident (MM/DD/YYYY HH:MM:SS AM/PM)
Primary Type	Primary crime category (e.g., THEFT, BATTERY, NARCOTICS)
Description	Sub-classification within the primary type

Column	Description
Location Description	Physical location type (e.g., STREET, RESIDENCE)
Arrest	Boolean — TRUE if an arrest was made
Domestic	Boolean — TRUE if classified as a domestic incident
District	Chicago Police Department district number
Community Area	Chicago community area number
Latitude / Longitude	WGS84 geographic coordinates
Year / Hour / Month / DayOfWeek	Temporal features extracted from the Date column

2.3 Scale and Infrastructure

At 7.8 million rows, standard single-threaded Pandas operations are impractically slow. The project uses two frameworks in tandem: Dask for the initial load, cleaning, and EDA (parallelised across CPU cores via lazy evaluation), and PySpark for the machine learning pipeline, where MLlib's cluster-aware implementations of scaling, clustering, and classification are far more efficient at this scale.

3. Methodology

3.1 Data Cleaning and Preprocessing

The following steps were applied to the raw CSV before any analysis:

- Deduplication: Rows sharing the same ID were dropped using per-partition deduplication in Dask, avoiding a costly cross-partition shuffle.
- Null removal: Any row containing at least one null value was dropped. Zero missing values remained after this step.
- Coordinate filtering: Records where Latitude, Longitude, X Coordinate, or Y Coordinate were exactly zero were excluded, these represent unresolved location data.
- Date parsing: The Date column was converted from string to datetime, enabling extraction of Hour, Month, DayOfWeek, and Year.
- Rare category consolidation: The top 25 most frequent values in Location Description and Description were retained; all others were labelled OTHER.
- Column pruning: Unused columns (Case Number, IUCR, Updated On, FBI Code, Beat, Ward) were dropped.

3.2 Exploratory Data Analysis

The EDA phase examined crime distribution across three dimensions: type, time, and geography, to surface patterns that would inform the modelling choices made later.

3.3 Feature Engineering for Clustering

The clustering feature vector used three dimensions: Latitude, Longitude, and Hour. This captures both where and when a crime occurred, without including features that would blur spatial coherence. Features were assembled with VectorAssembler and standardised with StandardScaler (std scaling, mean not centred), so geographic and temporal scales contribute equally to the distance metric.

3.4 Clustering: Bisecting K-Means

The clustering algorithm was Bisecting K-Means (BKM) from PySpark MLlib. BKM recursively splits the highest-variance cluster, making it far more efficient than standard K-Means at millions of records. It was configured with K = 30 clusters (fine-grained enough to capture distinct zones across the city) and a fixed seed of 42. Data was repartitioned into 200 Spark partitions before training.

3.5 Classification: Random Forest for Arrest Prediction

The supervised task was binary classification: predict whether an arrest was made (Arrest = True → label 1.0, False → label 0.0). Six features were used:

- typeIndex: Primary Type label-encoded via StringIndexer
- Hotspot_ID: cluster assignment from the BKM model (encodes geographic and temporal context compactly)
- Hour, DayOfWeek: time of the incident
- Latitude, Longitude: raw coordinates for fine-grained spatial signal

The model was a Random Forest Classifier with 100 trees, maximum depth 10, and 40 bins per feature. Data was split 80/20 (training / test), with the fixed seed 42 for reproducibility.

4. Exploratory Data Analysis — Results

4.1 Crime Type Distribution

Theft is the dominant crime type by a wide margin, followed by Battery and Criminal Damage. Together these three account for roughly half of all incidents. The distribution has a long tail, over a dozen other types each record hundreds of thousands of incidents, making Chicago's crime landscape genuinely diverse.

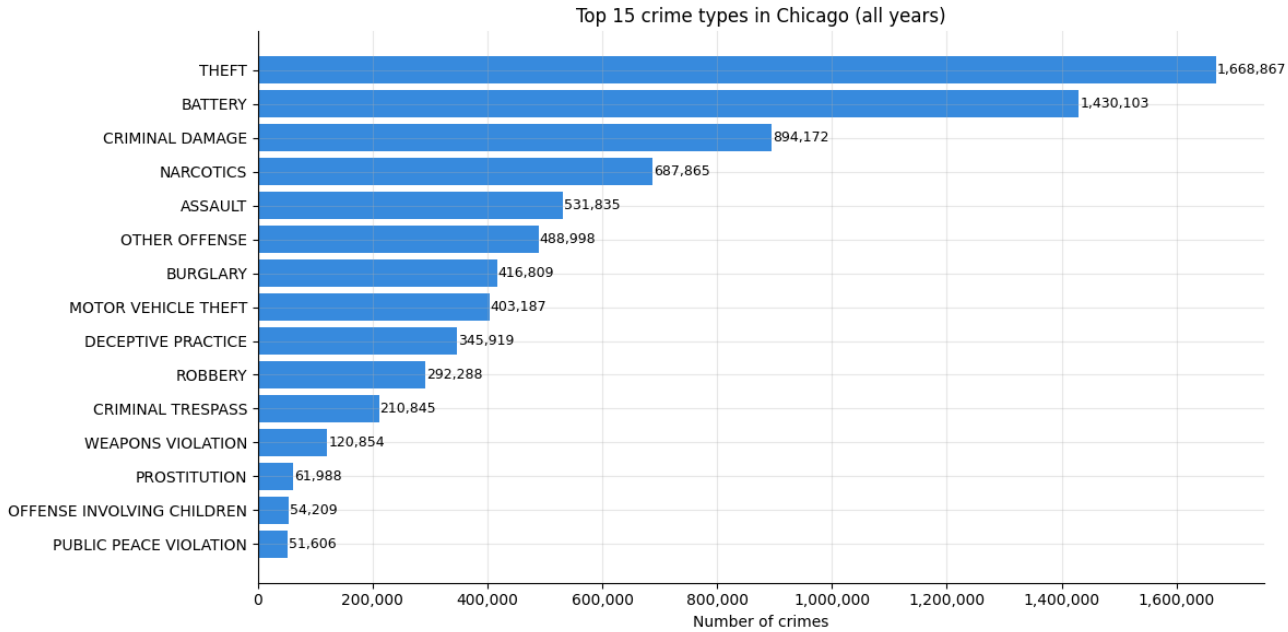

Figure 1: Top 15 crime types by total incident count (all years)

4.2 Location of Incidents

Streets are by far the most common crime location, which has real implications: outdoor public spaces are harder to secure than indoor ones, and street-level policing is the most direct lever available. Residences and apartments follow, representing a substantial domestic-context component handled very differently by enforcement.

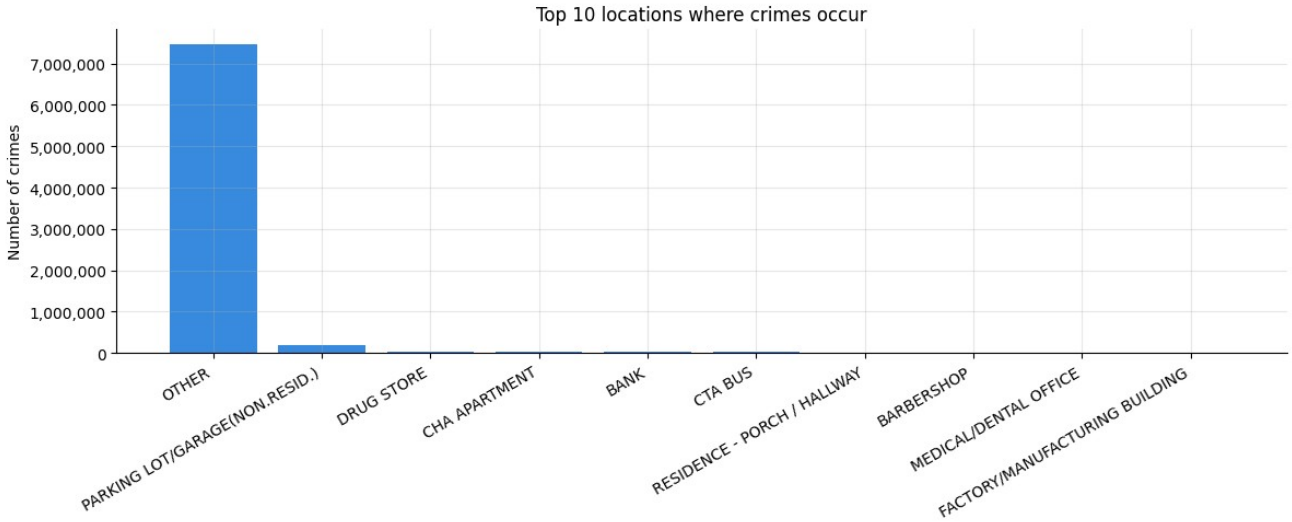

Figure 2: Top 10 location types where crimes are recorded

4.3 Temporal Patterns

4.3.1 Hour of Day

Crime follows a clear diurnal pattern: lowest around 4–5 AM, rising through the day, peaking at midnight. The midnight spike is partly a recording convention (incidents without an exact time logged as 00:00), but the elevated late-evening counts are genuine.

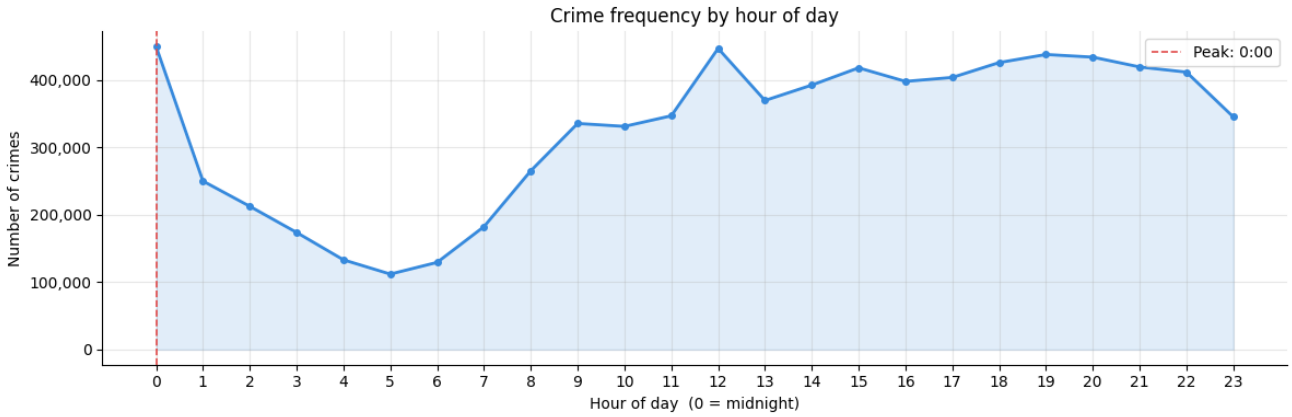

Figure 3: Crime frequency by hour of day (0 = midnight)

4.3.2 Day of Week

Crime is fairly evenly spread across days of the week. Friday is the peak, consistent with increased social activity as the working week ends, but between-day differences are modest.

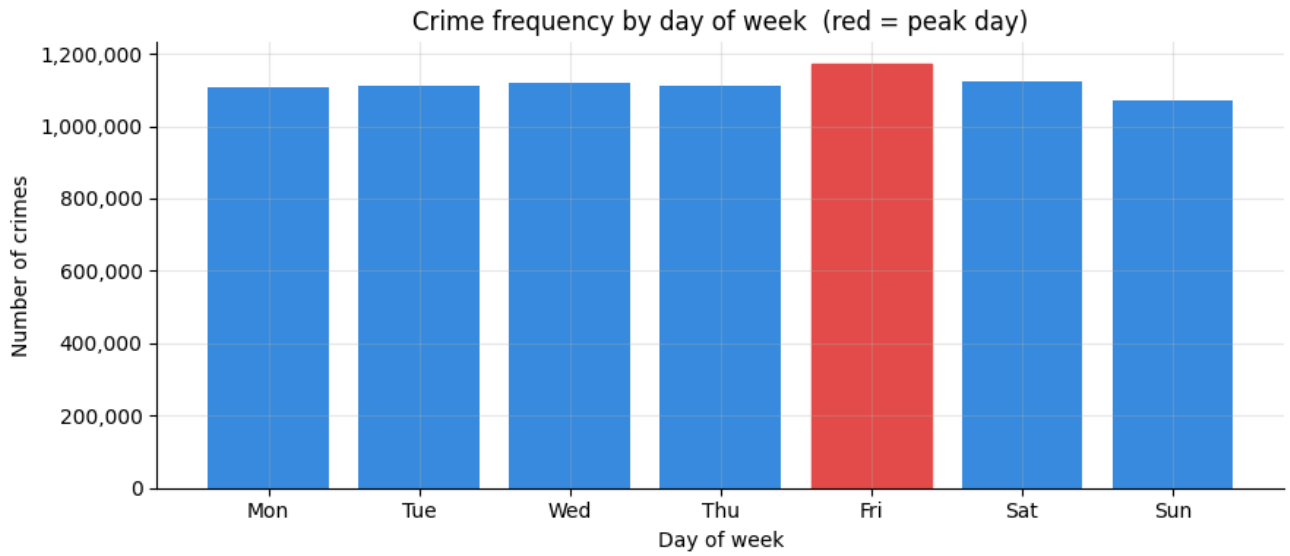

Figure 4: Crime frequency by day of week (red bar = peak, Friday)

4.3.3 Month and Season

July is the peak month and February the trough, a difference of roughly 30–40% in monthly counts. The warm-weather pattern is clear: more people outdoors means more opportunity for theft and assault. Seasonal resource planning is well supported by this data.

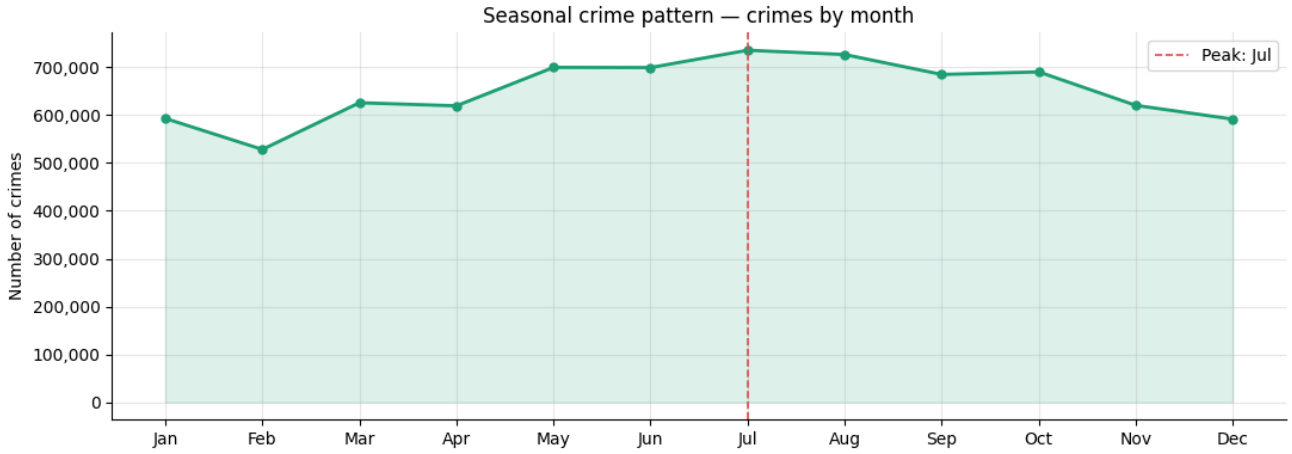

Figure 5: Total incidents by month across all years (red line = July peak)

4.3.4 Long-Term Trend (2003–2024)

Total recorded crime in Chicago has roughly halved since its early-2000s peak, from around 450,000 annual incidents to approximately 250,000 by 2024. This trend is sustained and broad-based, reflecting two decades of compounding improvements in policing, community programmes, and urban development.

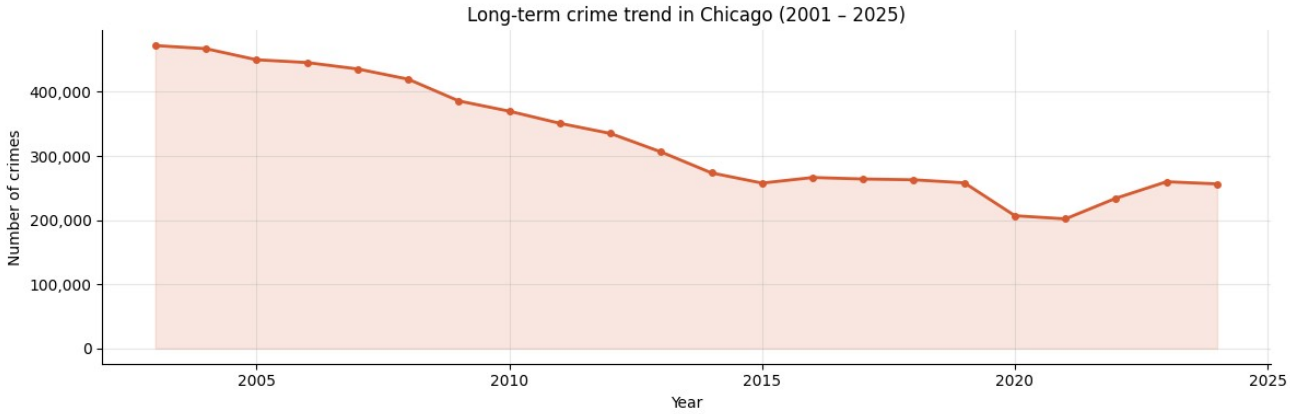

Figure 6: Long-term total crime trend in Chicago, 2003–2024

4.3.5 Top 5 Crime Type Trends

The decline is uneven across types. Narcotics dropped most sharply. Motor Vehicle Theft fell dramatically after 2007 then rebounded after 2014, mirroring national trends linked to keyless entry vulnerabilities. Theft and Battery declined more gradually.

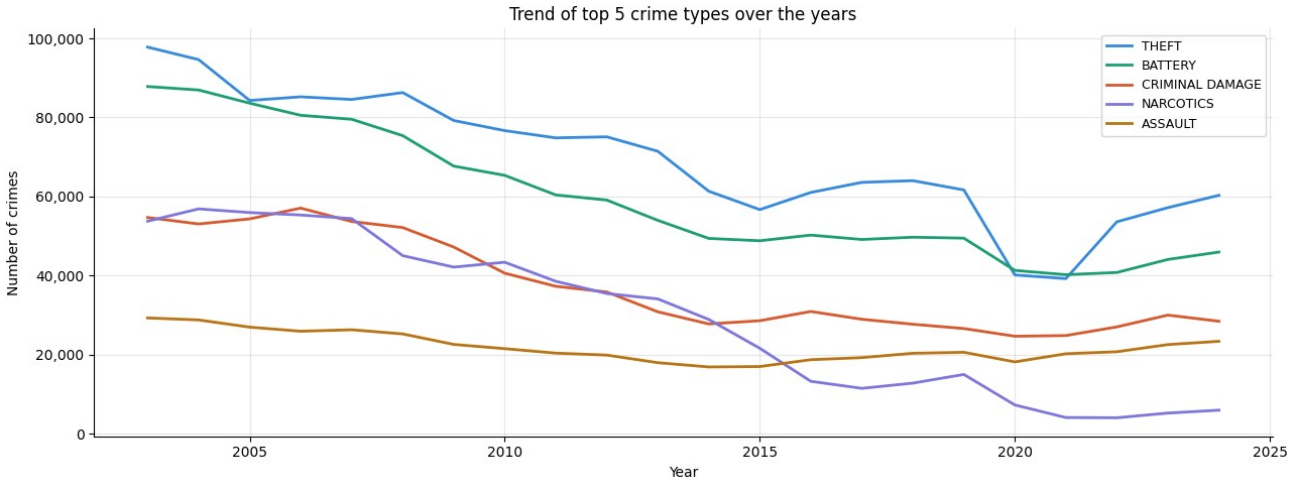

Figure 7: Annual trend for the top 5 crime types, 2003–2024

4.4 Hour vs. Day Heatmap

The heatmap puts hourly and daily patterns together in one view. The highest-density cells cluster in the late-night to early-morning window on Fridays and Saturdays. A visible midday concentration runs across all weekdays. The 4–6 AM window is consistently the lightest.

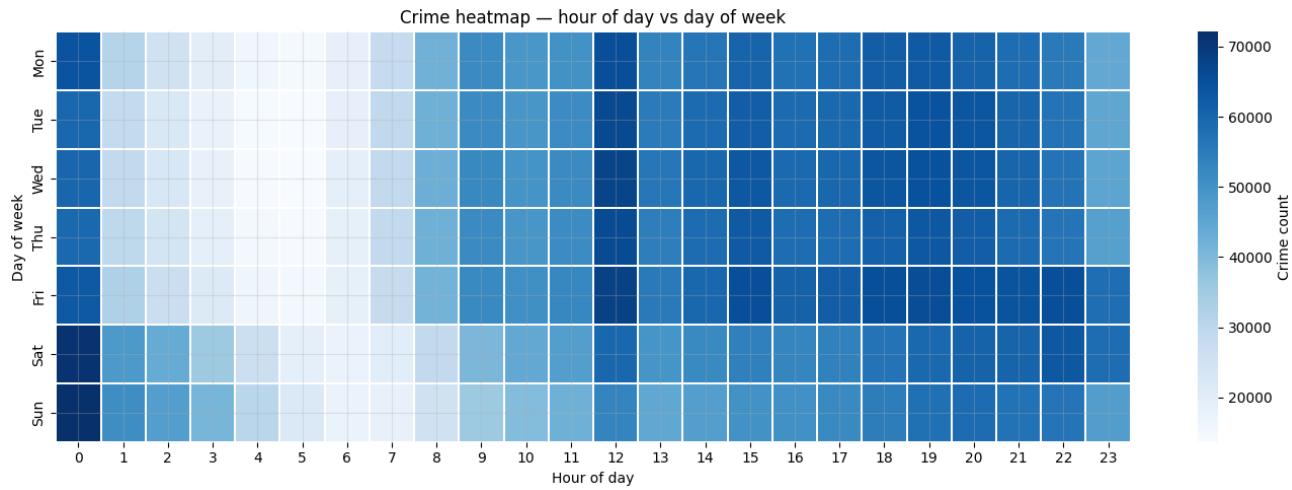

Figure 8: Crime density heatmap — hour of day vs. day of week (darker = more incidents)

4.5 Arrest Rates by Crime Type

Arrest rates vary enormously across crime types. Narcotics leads at 99.3%, most drug offences are detected proactively, so detection and arrest happen simultaneously. At the other end, Burglary (5.6%) and Criminal Damage (6.3%) sit well below 20%, as victims typically report these crimes after the perpetrator has already left. This variation is what makes the arrest-prediction model in Section 6 meaningful.

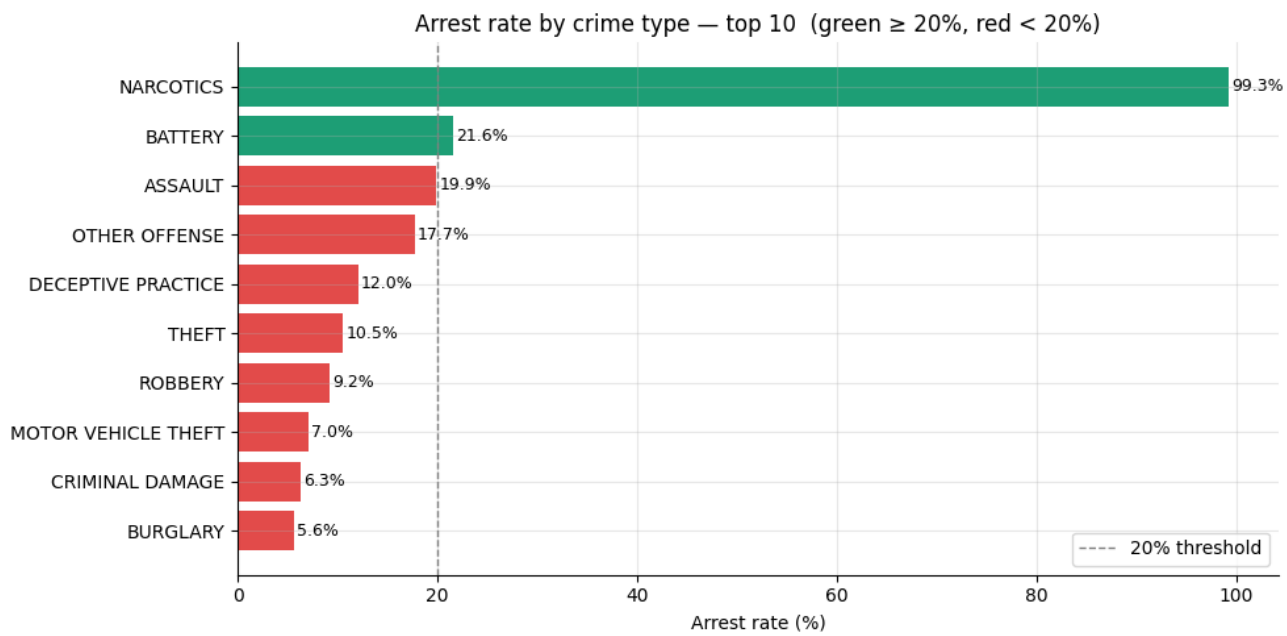

Figure 9: Arrest rate by crime type — top 10 (green ≥ 20%; red < 20%)

4.6 District-Level Distribution

Crime is not uniformly spread across Chicago's police districts. The most intensive district (highlighted red) handles a disproportionately large share of incidents. The clustering analysis in Section 5 goes deeper, identifying specific zones rather than working at the coarse district level.

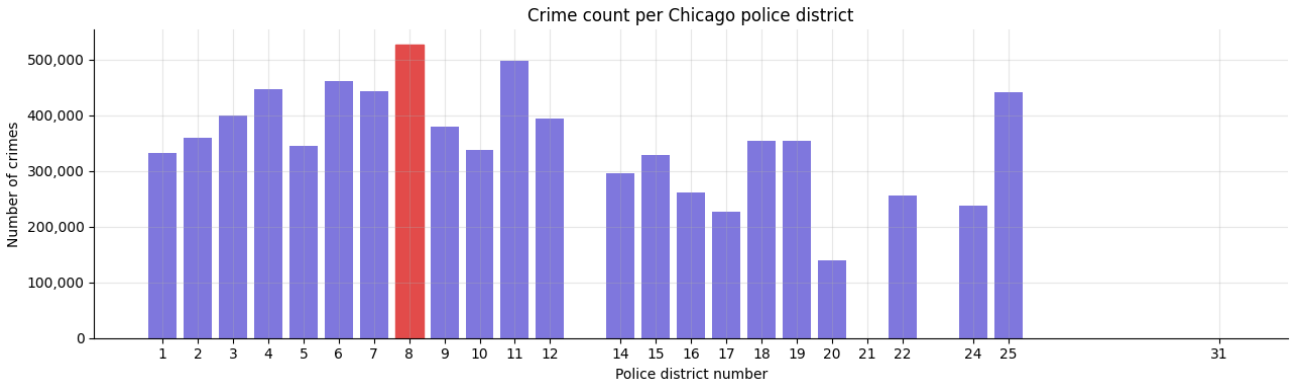

Figure 10: Total crime count per Chicago police district (red = highest volume)

4.7 Geographic Distribution

The scatter plot (50,000-point sample) maps the spatial footprint of crime across Chicago. The point cloud follows the street grid closely, with visibly higher density through the central and south-side areas, consistent with the district-level findings.

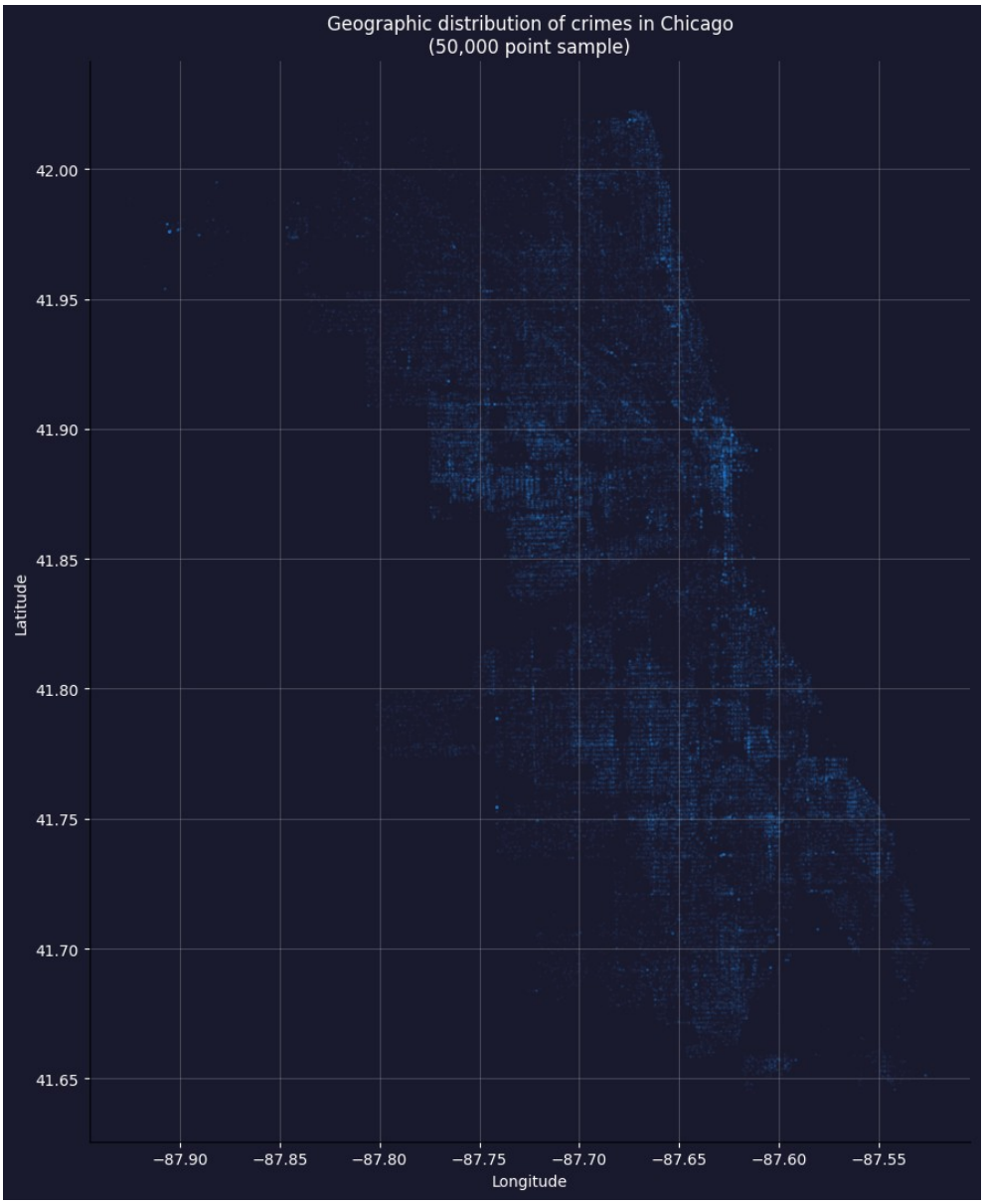

Figure 11: Geographic distribution of crimes — 50,000 point sample

5. Clustering for Hotspot Detection

The EDA established that crime is unevenly distributed across time and space. The clustering step quantifies this unevenness into 30 spatially and temporally coherent zones, each treatable as a distinct hotspot with its own crime profile.

5.1 Cluster Balance Check

The first validation checks how evenly the 7.9 million records are distributed across the 30 clusters. An uneven distribution is expected and desirable, clusters with counts well above the theoretical mean (total / 30) correspond to the densest crime zones. A perfectly uniform distribution would indicate the algorithm failed to find structure.

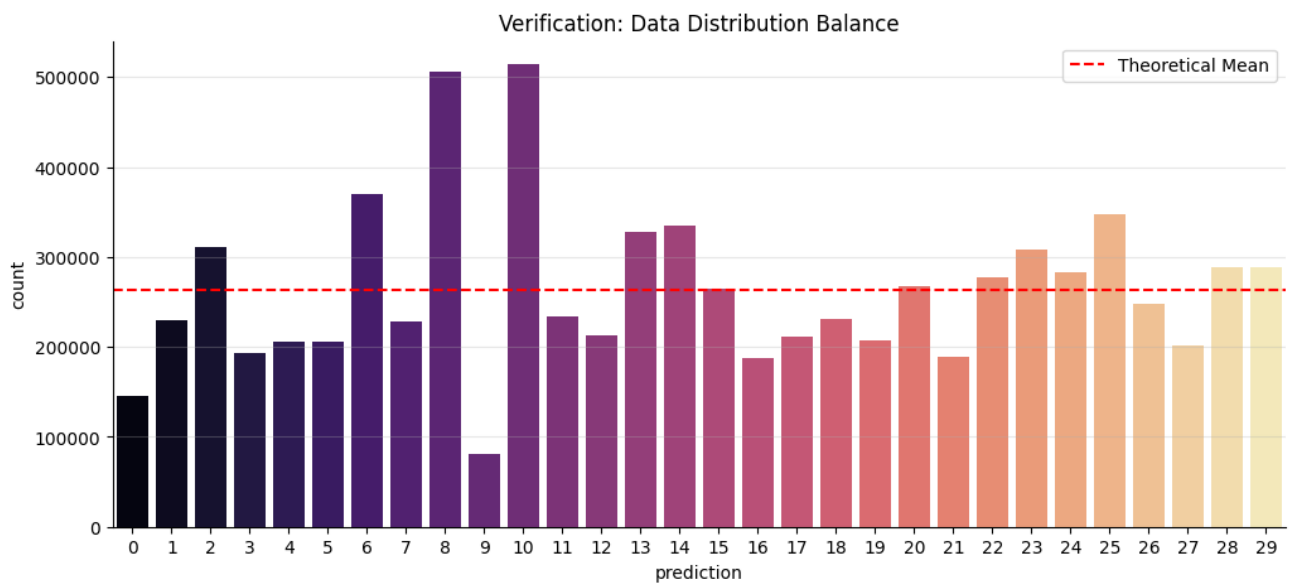

Figure 12: Crime records per cluster — red dashed line = theoretical equal-distribution mean

5.2 Temporal Coherence Check

The boxplot shows the distribution of the Hour feature within each cluster (1% random sample). Clusters with a narrow IQR are temporally specialised, crimes concentrate at specific hours. Clusters with a wide IQR are geographically coherent but time-agnostic. Both types are useful: specialised ones suggest natural patrol windows, mixed ones point to areas with persistent around-the-clock activity.

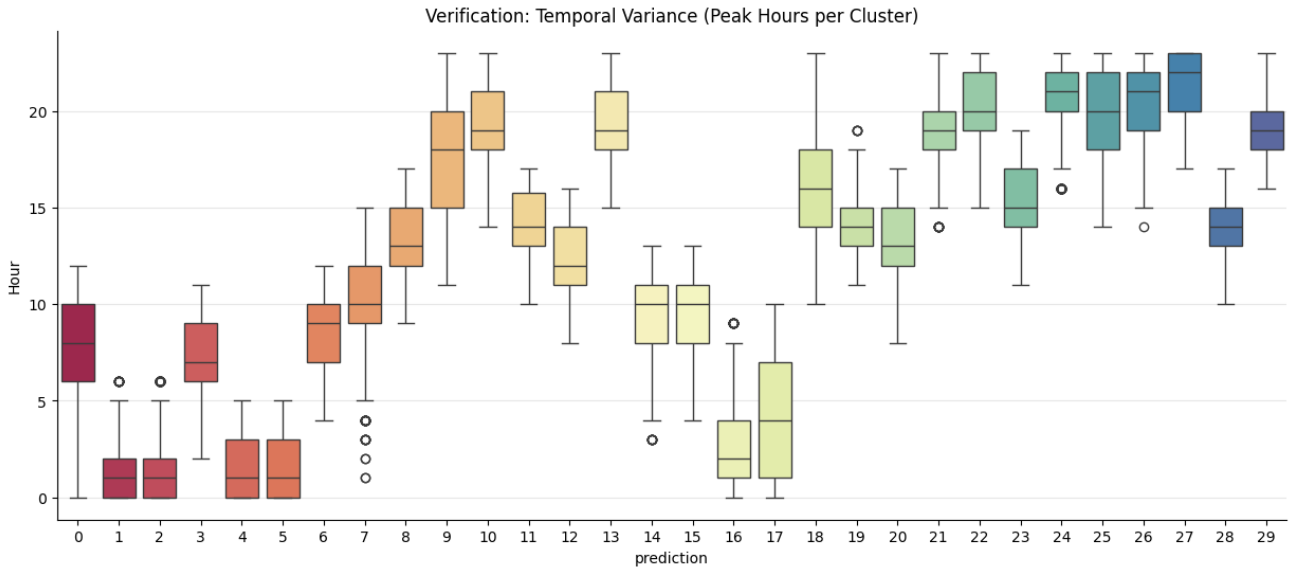

Figure 13: Distribution of crime hours per cluster — narrow IQR = temporally focused hotspot

5.3 Crime Type Signature per Cluster

The heatmap shows the crime type composition of each cluster, row-normalised so each cell reflects the share of that cluster's crimes belonging to each type. Two clusters with similar footprints can have completely different signatures, one primarily a theft zone, another a narcotics hotspot, which calls for different interventions even if they are geographically adjacent.

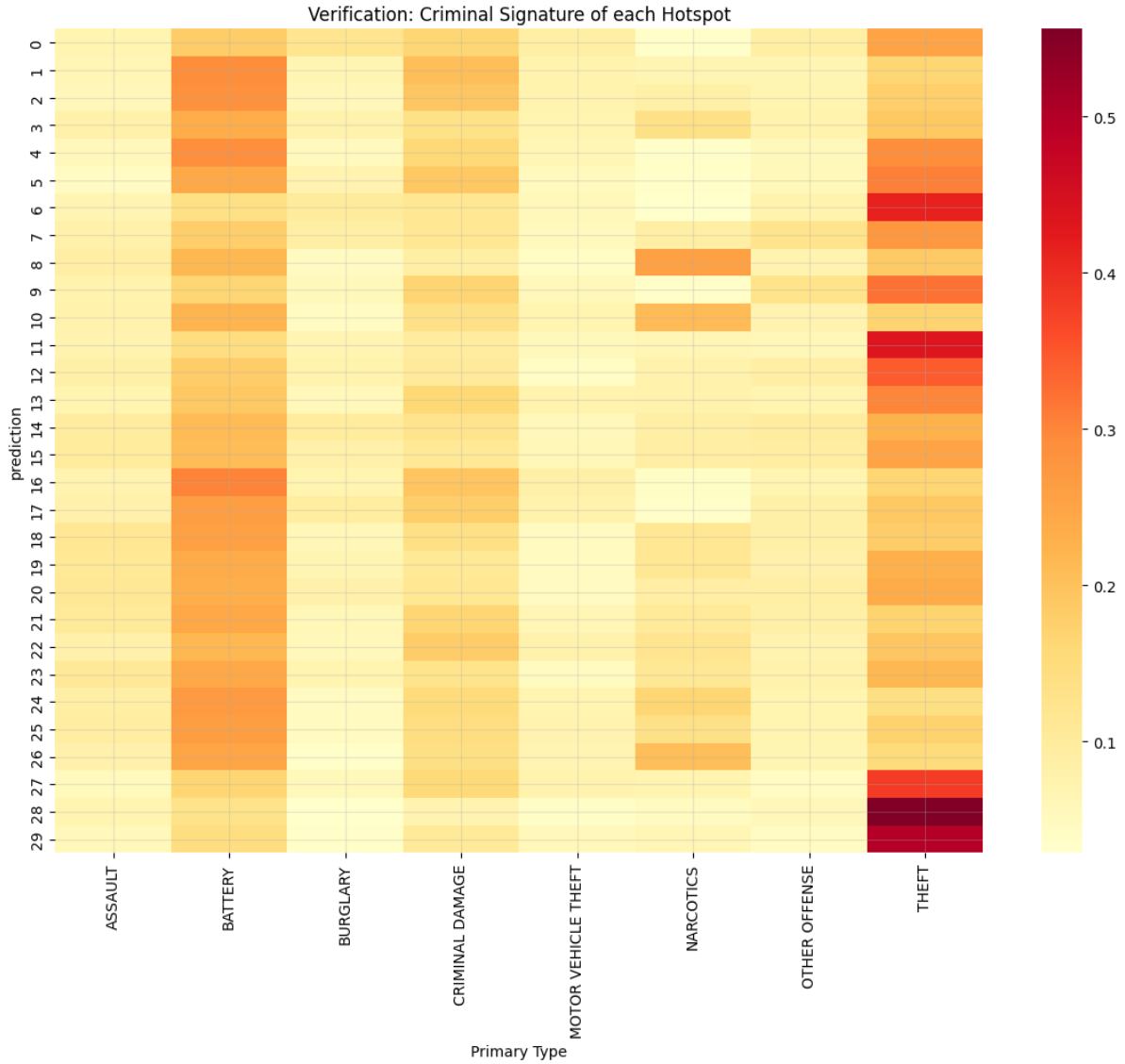

Figure 14: Row-normalised crime type composition per cluster — each row = one hotspot zone

5.4 Final Hotspot Map

Each point is a crime incident (1% sample), coloured by cluster assignment. White X markers show each cluster's centroid, the geographic and temporal centre of that hotspot. Clusters have clear spatial coherence: similarly coloured points group together, and the 30 centroids spread across the city in a pattern that tracks the crime density seen in Figure 11.

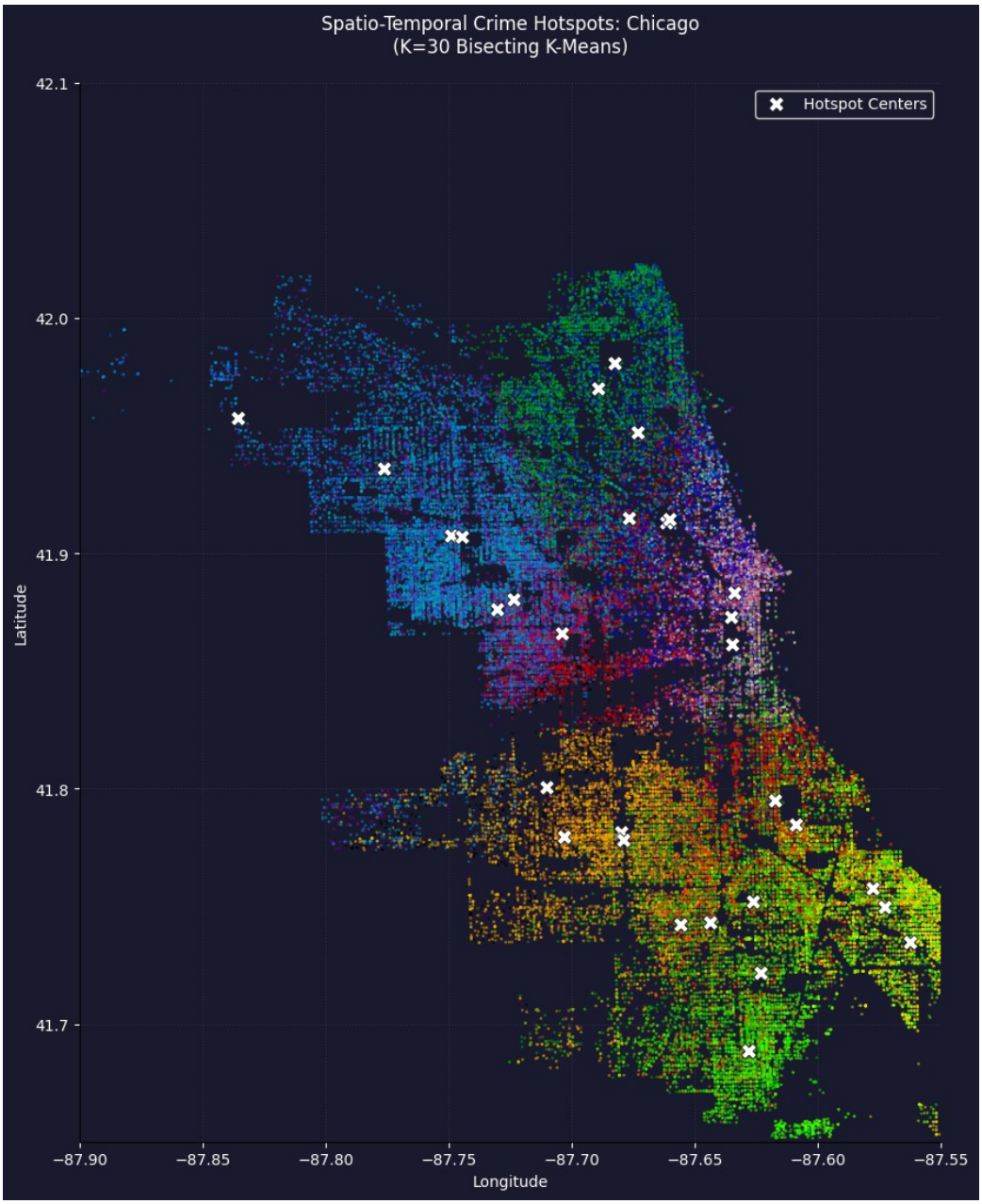

Figure 15: Spatio-temporal crime hotspot map — 30 Bisecting K-Means clusters, white X = centroids

6. Classification Model — Arrest Prediction

After clustering, the Hotspot_ID assigned to each incident was incorporated as a feature in a supervised classifier trained to predict whether an arrest would be made. If the model reliably identifies incidents that share characteristics with historically arrested cases, it can support investigative prioritisation and help analysts understand what drives arrest outcomes.

6.1 Feature Set and Model

Six features were assembled into the classifier:

- typeIndex: Primary Type label-encoded by Spark's StringIndexer. The most informative single feature, given the wide variation in arrest rates across crime types.
- Hotspot_ID: the BKM cluster assignment. Encodes both geographic and temporal context in one compact integer.
- Hour and DayOfWeek: time of incident.
- Latitude and Longitude: raw coordinates for fine-grained spatial signal beyond the cluster summary.

The Random Forest Classifier used 100 trees, depth 10, and 40 bins per feature. The 80/20 train-test split (seed 42) produced ~6.3M training records and ~1.58M test records.

6.2 Results

All metrics were computed on the held-out 20% test set:

Metric	Score	Interpretation
Area Under ROC (AUC)	0.8402	Strong ability to separate arrested vs. non-arrested cases
Weighted Precision	0.8746	When arrest was predicted, it was correct ~87.5% of the time
Weighted Recall	0.8652	Of all actual arrests, ~86.5% were correctly identified
F1 Score	0.8494	Balanced harmonic mean of precision and recall

An AUC of 0.84 means the model correctly ranks a randomly chosen arrested case above a randomly chosen non-arrested case 84% of the time. The F1 score of 0.85 confirms that precision and recall are both strong, the model is not simply predicting the majority class.

6.3 Confusion Matrix

Of the 1.58 million test records, the breakdown of predictions was:

		Predicted: No Arrest	Predicted: Arrest
Actual: No Arrest		**TN = 1,172,130**	FP = 13,755
Actual: Arrest		FN = 199,289	**TP = 195,522**

The false-negative count (199,289) is higher than false positives (13,755), reflecting the class imbalance: only 24.8% of incidents resulted in an arrest. The model is conservative, it avoids predicting arrests it is not confident about. This threshold could be adjusted depending on the intended use case.

6.4 Prediction Distribution

The model predicted no-arrest for 1,371,419 records (86.8%) and arrest for 209,277 (13.2%). This is below the actual arrest rate of 24.8%, confirming the conservative tendency, but the 195,522 true positives show the model is identifying genuinely arrest-likely incidents rather than randomly assigning predictions.

7. Key Findings

7.1 Crime Type and Arrest Patterns

- Theft is Chicago's most common crime but has one of the lowest arrest rates (10.5%). High volume combined with low enforcement effectiveness makes it a priority for prevention-focused strategies.
- Narcotics has a 99.3% arrest rate because offences are detected proactively, detection and arrest happen simultaneously. Its sharp long-term decline since the mid-2000s reflects sustained enforcement pressure.
- Motor Vehicle Theft partially rebounded after 2014 despite the broader downward trend, mirroring national patterns linked to keyless entry vulnerabilities.

7.2 Temporal Patterns

- Midnight is the statistical peak hour; 5 AM is the quietest. The genuine danger window runs roughly 9 PM to 1 AM.
- July is consistently the worst month; February the best. A 30–40% seasonal swing means resource planning should account explicitly for summer peaks.
- The late Friday to Saturday night window, visible in the heatmap as the darkest cells, is the most concentrated single crime window of the week.
- Day of week matters less than hour of day or month. Day-level scheduling adjustments alone would yield modest gains.

7.3 Geographic Concentration and Hotspots

- A small number of districts and clusters account for a disproportionate share of incidents. Uniform deployment of resources across the city systematically under-resources the highest-demand areas.
- Adjacent clusters can have completely different crime type signatures, requiring different interventions even in close geographic proximity.
- The 30 cluster centroids provide an updatable map of high-risk zones. As new data arrives, re-running the model tracks how hotspot boundaries shift over time.

7.4 Arrest Prediction Model

- The Random Forest classifier achieves AUC 0.84 and F1 0.85 on a 1.58M record test set, strong results given the 24.8% class imbalance.
- Including Hotspot_ID as a feature compresses geographic and temporal context into a single integer, letting the model learn that certain zones and times are strongly associated with arrest likelihood.

- The model's conservative threshold can be adjusted to trade precision against recall depending on the operational use case.

7.5 Long-Term Trend

- Chicago's overall crime rate has fallen roughly 50% since the early 2000s, a genuine, sustained improvement across all major crime types. Any current-day policy analysis should be framed against this long-run context.

Note: *This report covers the complete analytical pipeline i.e., data cleaning, exploratory analysis, geographic clustering (Bisecting K-Means, K=30), and arrest-prediction modelling (Random Forest). All work was conducted on the Chicago Crime Dataset (2001–present) using Dask for preprocessing and PySpark for machine learning.*